

Store Sales Forecasting

Data Warehouse and Analytics Project

1. Project Overview

This project builds a structured Data Warehouse for store sales data, enabling reliable reporting, trend analysis, and business decision-making. Raw sales records from a CSV file are cleaned, organized, and transformed into analytical views that answer key business questions about customers, products, and revenue.

The warehouse is built on Microsoft SQL Server using a three-layer medallion architecture:

Component	What Was Done	Outcome
Bronze Layer	Raw data loaded directly from the CSV source file	Unmodified source of truth preserved for audit
Silver Layer	Data cleaned, standardized, and enriched with derived fields	Analysis-ready, consistent, validated dataset
Gold Layer	Business-friendly views: dimensions and fact tables	Ready for dashboards, reporting tools, and analytics

2. Data Pipeline — How Data Flows

Data moves through a clear, sequential pipeline with checkpoints at each stage:

- Step 1 — Pre-processing (Databricks/PySpark): The raw CSV is first cleaned in Databricks before being loaded into SQL Server. Special characters (quotes, commas, hyphens, slashes) are removed from product names, and extra spaces are trimmed.
- Step 2 — Bronze Load: A stored procedure uses SQL Server's BULK INSERT to load the cleaned CSV into the bronze.ssf table. Each run truncates and reloads the table, with timing logged for monitoring.
- Step 3 — Silver Transformation: A second stored procedure reads from Bronze and writes to Silver. It strips prefixes from customer IDs and product IDs, and calculates a new unit_price column (sales divided by quantity).
- Step 4 — Gold Views: Three SQL views are created — dim_customers, dim_products, and fact_sales — forming a star schema optimized for reporting and joins.

3. Data Quality Assurance

A dedicated set of quality checks is run against the Bronze Layer before data is promoted to Silver. These ensure the data entering the warehouse is clean and reliable.

Check Type	What It Verifies
Duplicate / Null Primary Key	No two rows share the same row_id; no row_id is missing
Leading / Trailing Spaces	All text fields are trimmed — no invisible whitespace
Standardization Checks	Ship modes, segments, categories, regions are consistent
Negative / Null Numbers	Sales, quantity, discount, and profit have no negative or missing values
Invalid Date Order	Order date is never later than ship date

These checks are run as SQL queries that return zero rows if the data is clean. Any rows returned indicate issues that must be resolved before the Silver Layer is refreshed.

4. Key Data Fields

The dataset covers the following business information:

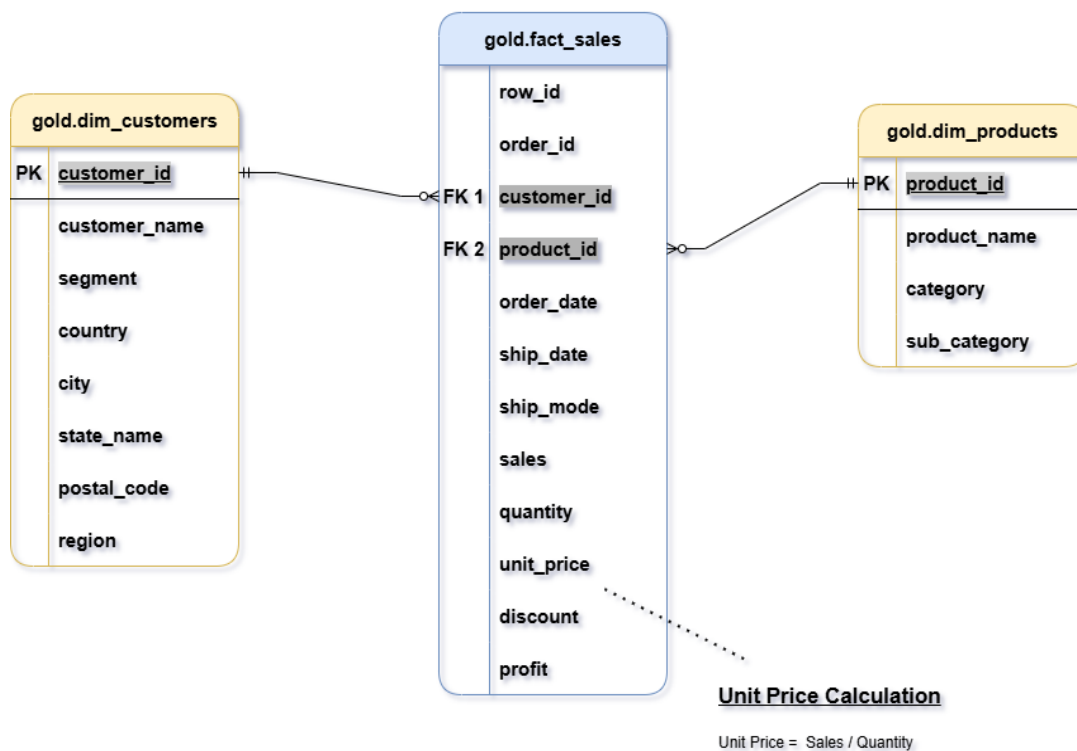
Field	Meaning
order_id	Unique identifier for each customer order
order_date / ship_date	When the order was placed and shipped
customer_id / name	Who placed the order (cleaned of prefix codes)
segment	Customer type: Consumer, Corporate, or Home Office
region / state / city	Geographic location of the customer
product_id / name	Which product was ordered (cleaned of prefix codes)
category / sub_category	Product classification (e.g., Furniture > Chairs)
sales	Revenue from the transaction
quantity	Number of units sold
unit_price	Derived: sales divided by quantity
discount	Discount applied (as a decimal, e.g., 0.2 = 20%)
profit	Profit earned on the transaction

5. Data Model

The Gold Layer follows a star schema — the standard structure for data warehouses. One central fact table holds sales transactions, joined to two-dimension tables that describe customers and products.

Table / View	Description
gold.dim_customers	One row per customer — name, segment, city, state, region, postal code
gold.dim_products	One row per product — name, category, sub-category
gold.fact_sales	All sales transactions — order date, ship date, sales, quantity, discount, profit, unit price

Star Schema



6. Analytics Capabilities

Once data reaches the Gold Layer, the following analyses are available out of the box:

6.1 Sales & Revenue Trends

- Monthly and yearly sales totals, quantities sold, and customer counts
- Running total of revenue over time (cumulative sales trend)
- Moving average of unit price across periods

6.2 Product Performance

- Top 5 and bottom 5 products by total revenue
- Revenue by product sub-category
- Average unit price and discount rate by sub-category
- Year-over-year product performance: is each product selling more or less than last year?
- Products compared against their own historical average — flagged as Above Avg., Avg., or Below Avg.

6.3 Customer Insights

- Top 10 customers by revenue
- Customer segmentation: VIP (>12 months history + >\$1,000 spend), Regular, and New
- Recency analysis: how many months since each customer last ordered
- Average Order Value (AOV) and Average Monthly Spend per customer

6.4 Geographic Distribution

- Customer counts by region, state, and city
- Volume of items sold by state and city

6.5 Part-to-Whole Analysis

- Percentage contribution of each sub-category to total sales revenue

6.6 Product Price Segmentation

- Products grouped into price bands: Below \$50, \$50–\$100, \$100–\$200, Above \$200
- Count of distinct products in each price band

6.7 Report Views

View	Description
gold.report_customers	Pre-built customer report with segments (VIP / Regular / New), recency, average spend
gold.report_products	Pre-built product report with performance segments, monthly revenue, average order value

Two pre-built report views (report_customers and report_products) are available for immediate use in dashboards or management reports without any further transformation.

7. Summary

This data warehouse project delivers a clean, well-structured, and analytically rich foundation for store sales reporting. Key achievements:

- Automated ETL pipeline from raw CSV to analytics-ready Gold Layer
- Star schema design for fast, simple reporting joins
- Pre-built customer and product report views ready for dashboard use
- Comprehensive analytics covering trends, ranking, segmentation, and YoY comparisons
- Built-in quality checks to ensure data reliability at every load

The warehouse is now ready to connect to a reporting tool such as Power BI or Tableau to deliver live dashboards to business stakeholders.